

# Problem 13

Approved and informed  
by discussions with  
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## König's Theorem: Why Interesting?

We know

$$\begin{aligned}
 p_{\text{tot}} &= \dot{E}_K \\
 \underbrace{p_{\text{ext}} + p_{\text{int}}}_{2 \text{ terms}} &= \underbrace{\dot{E}_{KG} + \dot{E}_{K/G}}_{2 \text{ terms, see 13 a, p.1}} \\
 &\quad \left( \sum \underline{F}_i^{\text{ext}} \right) \cdot \underline{v}_G \text{, see 13 b p.3}
 \end{aligned}$$

2 terms = 2 terms

It would be tempting to equate individual terms

$$\textcircled{1} \quad \underbrace{p_{\text{ext}} \stackrel{\checkmark}{=} \sum \underline{F}_i^{\text{ext}} \cdot \underline{v}_i}_{\text{def}} \stackrel{\times}{=} \left( \sum \underline{F}_i^{\text{ext}} \right) \cdot \underline{v}_G \stackrel{\checkmark}{=} \dot{E}_{KG}$$

see 13 d, p.7
see 13 b, p.3

$$\textcircled{2} \quad \underbrace{p_{\text{int}} \stackrel{\checkmark}{=} \sum \underline{F}_i^{\text{int}} \cdot \underline{v}_i}_{\text{def}} \stackrel{\times}{=} \dot{E}_{K/G}$$

see 13 c, p.4

but  $\checkmark$ 's are truths and  $\times$ 's are non-truths,  
as we will show in Problem 13 c and  
Problems 13.

See final page for Professor Ruina's summary.

Problem 13 a

Problem:

For a system of particles,

$E_k = \frac{1}{2} \sum m_i v_i^2 \Rightarrow E_k = \frac{1}{2} m_{tot} v_G^2 + \dots ?$

def:  $E_k = \frac{1}{2} \sum m_i v_i^2$

$$\Rightarrow E_k = \frac{1}{2} \sum m_i (\underline{v}_i \cdot \underline{v}_i)$$

$\underline{v}_i = \underline{v}_{i/G} + \underline{v}_G$

$$\Rightarrow E_k = \frac{1}{2} \sum m_i (\underline{v}_{i/G} + \underline{v}_G) \cdot (\underline{v}_{i/G} + \underline{v}_G)$$

$$\Rightarrow E_k = \frac{1}{2} \sum m_i (v_G^2 + v_{i/G}^2 + 2(\underline{v}_G \cdot \underline{v}_{i/G}))$$

$$\Rightarrow E_k = \frac{1}{2} \sum m_i \underbrace{v_G^2}_{\text{independent of } i} + \frac{1}{2} \sum m_i v_{i/G}^2 + \sum m_i (\underline{v}_G \cdot \underline{v}_{i/G})$$

$$\Rightarrow E_k = \frac{1}{2} (\underbrace{\sum m_i}_{\text{definition of } m_{tot}}) v_G^2 + \frac{1}{2} \sum m_i v_{i/G}^2 + \underline{v}_G \cdot \underbrace{\sum m_i \underline{v}_{i/G}}_{=0, \text{ see aside } \textcircled{1} \text{ on next page}} \quad \textcircled{i}$$

Conclude:  $E_k = \frac{1}{2} m_{tot} v_G^2 + \frac{1}{2} \sum m_i v_{i/G}^2$

 $E_{kG}$ 
 $E_{k/G}, \text{ rest of expression}$

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## Problem 13a, continued

① Aside

$$\text{begin with } \sum m_i \underline{r}_i / G = \underbrace{\sum m_i \underline{r}_i}_{\text{by definition}} - \underbrace{\sum m_i \underline{r}_G}_{\text{ii}} = \underline{0}$$

 $= m_{\text{tot}} \underline{r}_G$   
by definition

 $= m_{\text{tot}} \underline{r}_G$ , see  
② below

$$\Rightarrow \sum m_i \underline{v}_i / G = \frac{d}{dt} (\sum m_i \underline{r}_i / G) = \underline{0}, \text{ return to ①, p. 2}$$

$$\text{② } \sum m_i \underline{r}_G = \underline{r}_G \sum m_i = \underline{r}_G m_{\text{tot}}, \text{ return to ii above}$$

Problem 13 b

Problem:

For a system of particles,

$$(\sum \underline{F}_i^{\text{ext}}) \cdot \underline{v}_G \stackrel{?}{=} \frac{d}{dt} \left( \frac{1}{2} m_{\text{tot}} v_G^2 \right)$$

$$(\sum \underline{F}_i^{\text{ext}}) \cdot \underline{v}_G = \underbrace{m_{\text{tot}} \underline{a}_G}_{\textcircled{1} \text{ aside below}} \cdot \underline{v}_G \quad \textcircled{i}$$

$$\underline{\underline{\text{and}}} \quad \underline{\underline{\frac{d}{dt} \left( \frac{1}{2} m_{\text{tot}} v_G^2 \right)}} = \frac{1}{2} m_{\text{tot}} \underbrace{\frac{d}{dt} (\underline{v}_G \cdot \underline{v}_G)}_{\textcircled{ii}, \text{ see } \textcircled{2} \text{ aside below}} = m_{\text{tot}} \underline{a}_G \cdot \underline{v}_G$$

Conclude:  $(\sum \underline{F}_i^{\text{ext}}) \cdot \underline{v}_G = \frac{d}{dt} \left( \frac{1}{2} m_{\text{tot}} v_G^2 \right)$  always true

aside  $\textcircled{1}$   $\underline{F}^{\text{tot}} = \underbrace{\sum \underline{F}_i^{\text{ext}}}_{\text{def}} + \sum \underline{F}_i^{\text{int}}, \quad \sum \underline{F}_i^{\text{int}} = \underline{0}$  [force units]

by the axiom of mechanics that internal forces and

moments sum to zero (see Ruina and Pratap 2015 p. 1038)

$$\Rightarrow \underline{F}^{\text{tot}} = \sum \underline{F}_i^{\text{ext}}$$

with LMB ( $\underline{F}^{\text{tot}} = m \underline{a}_G$ ),  $\sum \underline{F}^{\text{ext}} = m \underline{a}_G$ , return to  $\textcircled{i}$  above

aside  $\textcircled{2}$   $\frac{d}{dt} (\underline{v}_G \cdot \underline{v}_G) = \underbrace{\frac{d}{dt} \underline{v}_G}_{\underline{a}_G} \cdot \underline{v}_G + \underline{v}_G \cdot \underbrace{\frac{d \underline{v}_G}{dt}}_{\underline{a}_G} = 2 \underline{a}_G \cdot \underline{v}_G$

return to  $\textcircled{ii}$  above



# Problem 13c

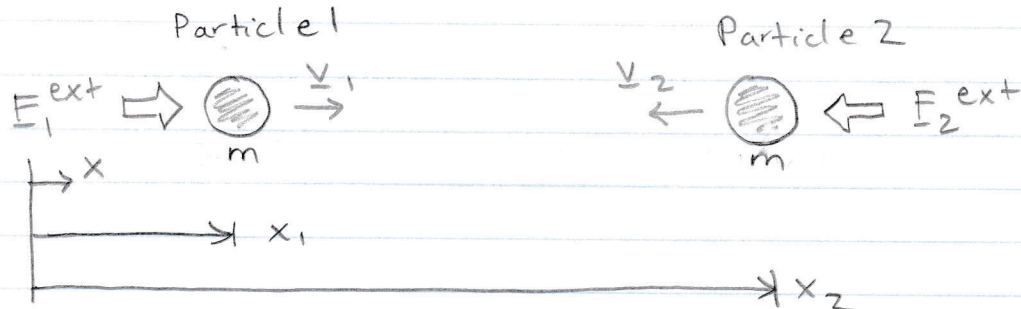
I. Problem: For a system of particles,

$$\underbrace{p_{int}}_{\text{LHS}} \stackrel{?}{=} \underbrace{\frac{d}{dt} \left( E_{K/G} = \frac{1}{2} \sum m_i v_{i/G}^2 \right)}_{\substack{\text{from 13a} \\ \text{RHS}}}$$

II. Show  $\text{LHS} \neq \text{RHS}$  for following counter-example

2 point masses ( $m$ ) in rectilinear motion with no internal forces and setup in diagrams and FBDs below

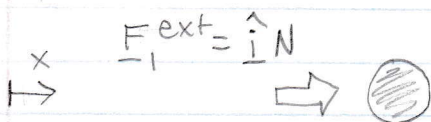
A. Diagram for Counter-example



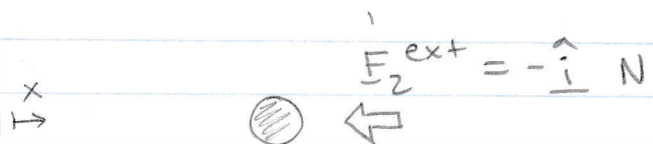
parameters:  $m = 1 \text{ kg}$ ,  $F_1^{\text{ext}} = -F_2^{\text{ext}} = 1 \hat{i} \text{ N}$

initial conditions:  $v_1(0) = 4 \hat{i} \text{ m/s}$ ,  $v_2(0) = -2 \hat{i} \text{ m/s}$

FBD of Particle 1



FBD of Particle 2



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Problem 13c, continued

B. evaluate LHS and RHS from problem statement for counter-example

LHS: no internal forces assumed  $\Rightarrow p^{int} = \sum \underbrace{\underline{F}_i^{int} \cdot \underline{v}_i}_{=0 \forall i} = 0 \frac{Nm}{s}$

def

RHS:  $\frac{d}{dt} \left( \frac{1}{2} \sum \underbrace{m_i}_{=1kg} \underline{v}_i / G^2 \right) = \frac{d}{dt} \left( \frac{1}{2} (\underbrace{v_1/G^2}_{=t^2+6t+9} + \underbrace{v_2/G^2}_{=t^2-6t+9}) \right)$

①, See Aside ① on next page

$$\Rightarrow \frac{d}{dt} \left( \frac{1}{2} \sum m_i \underline{v}_i / G^2 \right) = \frac{d}{dt} (t^2 + 9) = 2t \frac{Nm}{s}$$

$$\Rightarrow \frac{d}{dt} \left( \frac{1}{2} \sum m_i \underline{v}_i / G^2 \right) \neq 0 \frac{Nm}{s} \forall t \Rightarrow LHS \neq RHS$$

Conclude  $p^{int} \stackrel{*}{=} \frac{d}{dt} \left( \frac{1}{2} \sum m_i \underline{v}_i / G^2 \right)$

not equivalent

## Problem 13c, continued

Aside ①, find  $v_{1/G}^2$  and  $v_{2/G}^2 \Rightarrow$  use LMB to find accelerations then integrate to find velocities

LMB Particle 1:  $\underbrace{F_1^{\text{ext}}}_{=1\hat{i}\text{ N}} = m_1 \underbrace{a_1}_{=1\text{ kg}} \Rightarrow \dot{\underline{v}}_1 = \hat{i} \text{ m/s}$

$$\int_0^t \dot{\underline{v}}_1 dt' = \int_0^t \hat{i} dt' \Rightarrow \underline{v}_1 = (t+4) \hat{i} = \underline{v}_1(0)$$

LMB Particle 2:  $\underbrace{F_2^{\text{ext}}}_{=-1\hat{i}\text{ N}} = m_2 \underbrace{a_2}_{=1\text{ kg}} \Rightarrow \dot{\underline{v}}_2 = -\hat{i} \text{ m/s}$

$$\int_0^t \dot{\underline{v}}_2 dt' = \int_0^t -\hat{i} dt' \Rightarrow \underline{v}_2 = (-t-2) \hat{i} = \underline{v}_2(0)$$

also need  $\underline{v}_G$ :  $\underline{v}_G = (\sum m_i \underline{v}_i) / m_{\text{tot}} = m(\underline{v}_1 + \underline{v}_2) / (2m) = \hat{i} \frac{m}{s}$   
 $\quad \quad \quad \text{def.} \quad \quad \quad = (t+4)\hat{i} = (-t-2)\hat{i}$

Find  $v_{1/G}^2$ :  $\underline{v}_{1/G} = \underline{v}_1 - \underline{v}_G = (t+4)\hat{i} - \hat{i} = (t+3)\hat{i} \frac{m}{s}$   
 $\Rightarrow v_{1/G}^2 = (t+3)^2 \frac{m^2}{s^2} = t^2 + 6t + 9 \frac{m^2}{s^2}$

Find  $v_{2/G}^2$ :  $\underline{v}_{2/G} = \underline{v}_2 - \underline{v}_G = (-t-2)\hat{i} - \hat{i} = (-t-3)\hat{i} \frac{m}{s}$   
 $\Rightarrow v_{2/G}^2 = (-t-3)^2 \frac{m^2}{s^2} = t^2 - 6t + 9 \frac{m^2}{s^2}$

Return to (i) on p.5 if Problem 13c  
 (ii) on p.7 if Problem 13d



Part 13d (Bonus to finish showing "Why Interesting?")

I. Problem: For a system of particles,

$$\underbrace{(p_{\text{ext}} = \sum \underline{F}_i^{\text{ext}} \cdot \underline{v}_i)}_{\text{def, LHS}} \stackrel{?}{=} \underbrace{(\sum \underline{F}_i^{\text{ext}}) \cdot \underline{v}_G}_{\text{RHS}}$$

II. show  $\text{LHS} \neq \text{RHS}$  for counter-example in 13c

$$\begin{aligned} \text{LHS: } p_{\text{ext}} = \sum \underline{F}_i^{\text{ext}} \cdot \underline{v}_i &= \underline{F}_1^{\text{ext}} \cdot \underline{v}_1 + \underline{F}_2^{\text{ext}} \cdot \underline{v}_2 = 2t + 6 \frac{\text{Nm}}{\text{s}} \\ &= \hat{i} N \parallel = (t+4) \hat{i} N \parallel = -\hat{i} N \parallel = (-t-2) \hat{i} N \end{aligned}$$

See Problem 13c, Aside ① on p. 6

$$\begin{aligned} \text{RHS: } (\sum \underline{F}_i^{\text{ext}}) \cdot \underline{v}_G &= (\underbrace{\underline{F}_1^{\text{ext}}}_{=\hat{i} N} + \underbrace{\underline{F}_2^{\text{ext}}}_{=-\hat{i} N}) \cdot \underbrace{\underline{v}_G}_{=\hat{i} \frac{\text{m}}{\text{s}}} = 0 \frac{\text{Nm}}{\text{s}} \end{aligned}$$

②, See Problem 13c, Aside ① on p. 6

$\text{LHS} \neq \text{RHS} \Rightarrow$

Conclude  $p_{\text{ext}} = \sum \underline{F}_i^{\text{ext}} \cdot \underline{v}_i \not\equiv (\sum \underline{F}_i^{\text{ext}}) \cdot \underline{v}_G$   
 not equivalent



Professor Ruina's Oct. 12 posted solution, which he asked to be included.

Total Power into system

$$= \dot{P}^{\text{tot}}$$

$$= \boxed{\dot{P}^{\text{ext}}} + \boxed{\dot{P}^{\text{int}}}$$

$$= \sum \vec{F}_i^{\text{ext}} \cdot \vec{V}_i + \sum \vec{F}_i^{\text{int}} \cdot \vec{V}_i$$

BUT

$$\dot{P}^{\text{int}} \neq \dot{E}_{K/G}$$

and

$$\dot{P}^{\text{ext}} \neq \dot{E}_{K/G}$$

Rate of Change of System Kinetic Energy

$$= \dot{\vec{E}}_K$$

$$= \boxed{\dot{\vec{E}}_{K/G}} + \boxed{\dot{\vec{E}}_{K/G}}$$

$$= \frac{d}{dt} \left( \frac{m_{\text{tot}}}{2} V_G^2 \right) + \frac{d}{dt} \left( \frac{1}{2} \sum m_i \vec{V}_i \cdot \vec{V}_i \right)$$

$$= \left( \sum \vec{F}_i^{\text{ext}} \right) \cdot \vec{V}_G + \sum m_i \vec{V}_i \cdot \frac{d\vec{V}_i}{dt}$$

EX) 1D

No internal forces.

$\dot{P}^{\text{int}} = 0$  but  $\dot{E}_{K/G} > 0$

$\dot{P}^{\text{ext}} > 0$  but  $\dot{E}_{K/G} = 0$

System has 2 masses